



Ph.D. Positions in Trustworthy and Multi-Agent GeoAI

Group of Dr. Peng Luo · Incoming Assistant Professor, August 2026

School of Earth, Environment, and Sustainability · University of Iowa

I will join the School of Earth, Environment, and Sustainability (SEES) at the University of Iowa as a tenure-track Assistant Professor in August 2026, and I am building a new research group at the intersection of Geographic Information Science (GIScience), Geospatial Artificial Intelligence (GeoAI), and spatial data science. I am recruiting **two fully funded Ph.D. students** to begin in **Fall 2027**. Each position is fully funded through a research or teaching assistantship, covering stipend, tuition, and benefits.

Research Directions

Position 1 — Trustworthy GeoAI. Developing reliable, interpretable, and uncertainty-aware geospatial AI. Topics include spatial explicit modeling, geospatial explainable AI (XAI), and uncertainty quantification of GeoAI models for spatial prediction and association analysis.

Position 2 — Multi-Agent GeoAI Modeling. Building LLM-driven, multi-agent frameworks for automated geospatial model discovery, reasoning, and scalable, self-evolving spatial analysis.

Qualifications

- A Bachelor's or Master's degree in GIScience, computer science, statistics, remote sensing, geography, or a closely related field.
- Strong programming skills (e.g., Python) and a genuine interest in spatial data science and AI.
- Self-motivated, curious, and able to communicate clearly in written and spoken English. Prior research experience is a plus but is not required.

What I Offer

- Full funding through a research or teaching assistantship (stipend, tuition, and benefits) for the duration of the Ph.D.
- Close, individualized mentorship and a clear path toward an academic or research career.
- An active international collaboration network and support for conference travel, computing resources, and publications.

Mentoring Philosophy

My mentoring philosophy rests on a simple Kantian principle: always treat people as ends in themselves, never merely as means. From that starting point, here is what I hope for in a student, in order of priority. My foremost responsibility is not to maximize your output or achievements, but to provide the opportunity and the space for you to experience the journey of a Ph.D. and come to know yourself on your own terms.

On that foundation, I hold intellectual freedom as paramount. I hope you will find the geographic questions that truly fascinate you, and — in the best case — develop enough healthy intellectual audacity to believe that you are the one who cares about, and can solve, a particular problem.

I will also be honest about my third priority: as a junior faculty member, I face real pressures of survival and promotion. So, while the first two come first, I hope your interests can also connect with my own research trajectory. When they do, I can offer the most expert guidance with greater confidence, and we both thrive. I see these three not as competing but as reinforcing one another: the students who do their most original work are usually those given the freedom to care deeply about something of their own.

How to Reach Me

If you are interested, please **email me directly** at **peng-luo@uiowa.edu** with your CV, transcripts, and a brief statement of research interests (indicate which position interests you), using the subject line “Prospective Ph.D. — [your name].” I review inquiries on a rolling basis and will follow up with promising candidates for a conversation. I encourage reaching out before submitting a formal application.

Formal application: Admitted students enroll through the Ph.D. in Geography program (GIScience track) at SEES. The University's application deadline is **December 1** for the following Fall. I will guide selected candidates through the process — please contact me first.

Dr. Peng Luo · peng-luo.com · peng-luo@uiowa.edu

School of Earth, Environment, and Sustainability, University of Iowa, Iowa City, IA 52242